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Examiners' Report Summer 2026

Pearson Edexcel GCE

In AS Mathematics (8MA0)

Paper 22 Mechanics

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General Comments

Most learners were able to attempt all four questions, and time did not appear to be a limiting factor.

Where a printed answer is given, as in 3(b), learners should include enough detail in their working to justify the award of all available marks.

There were some excellent scripts; however, in a substantial number of responses, the standard of presentation made it more difficult for examiners to follow the working and award marks appropriately.

As stated on the front of the question paper, learners should show sufficient working to make their methods clear to the examiner. Correct answers without supporting working may not score all, or any, of the available marks.

If a learner runs out of space for an answer, they are advised to use a supplementary sheet.

Comments on Individual Questions

Question 1

This question provided an accessible start to the paper but there were some learners who made errors, especially in part (b).

Part (a)

This question part was answered well by most learners. Correct answers were evenly split between surd and decimal forms. Of those responses that scored just one method mark, most obtained the mark for finding a magnitude rather than using either $F = ma$ or its vector equivalent.

Part (b)

This part was more challenging, although for those responses who were able to successfully answer part (a), most were able to find the correct distance travelled. However, a significant number of responses stated or used $s = at$ rather than using the correct $suvat$ formula for distance.

Question 2

Part (a)

Most learners drew a speed-time graph of the correct shape for the motion of the car between points A and C. Occasionally, the constant deceleration phase was drawn as a curve, but such instances were rare. Some started their graph before point A, either with a constant speed section or an acceleration section starting from the origin. A constant speed section before A did not affect the required shape of the graph, whereas acceleration from the origin led to an incorrect trapezium.

The second mark required correctly labelled axes with the relevant values. Common errors included labelling T twice on the t-axis, rather than T and $2T$, and omitting 24 from the speed axis. Placing T in the middle of each section was accepted where correct delineators were also shown.

Part (b)

This part was completed with varying degrees of success. The most common approach was to equate the area of the trapezium to the given distance to produce an equation in T only. This was generally done well, with many correct responses seen for the value of T. The distance was given in kilometres, so a conversion to metres was required. In some responses, this conversion was either omitted or carried out incorrectly.

Once a value for T *had* been found, most divided 24 by T to calculate the deceleration for the final phase. Many responses correctly found the acceleration to be -0.6 ms^{-2} , Since the question asked for the deceleration, the final mark was awarded only where this value was given as positive.

A significant minority of responses attempted to use *suvat* for the whole motion; this was not a valid method for the motion described and did not gain credit. A small number used an alternative valid approach by identifying the area of the second section of the graph as one third of 1440, before using *suvat* to find the deceleration.

Question 3

Part (a)

Although most learners wrote down the value of B correctly as 200, a few responses identified it incorrectly as 8, forgetting about the factor $\frac{1}{25}$ in the given expression for *h*.

Part (b)

It was given that the drone reached its maximum height at $t = 2$. Those learners who realised that differentiation of h was required tended to produce a correct derivative and $t = 2$ substitution to show $A = 60$, although again there was occasional confusion with the factor $\frac{1}{25}$. A fair number of responses only substituted $t = 2$ into the h expression, resulting in no valid progress.

Part (c)

The most common approach to find the value of T, the time when the drone landed, was to set the derivative of h equal to zero and solve for t. Alternatively, some responses started by setting $h = 0$ and solving for t. The use of a calculator was acceptable for the solution of both the quadratic and the cubic equation. The question required verification that **both** speed and height were zero at the point of landing. Either both equations needed to be solved, and the correct common root identified, or else the solution $T = 10$ from one equation had to be seen correctly substituted into the other in order to verify. Many of those responses who attempted this part of the question achieved two out of the three available marks through only considering one condition.

Question 4

Part (a)

This question part was answered well. Most learners were able to write down an appropriate equation of motion for each of the balls A and B, although a few responses seem to have them moving in opposite directions. Occasionally, additional "m's" were included in the working, and a few were only able to give the equation of motion for the whole system. A significant minority of responses were able to give the correct equations of motion for A and B, but did not solve the two simultaneous equations correctly for the values for a and T. A few obtained $T = 28$ (N) but then doubled it and stated that the tension was 56 (N).

Part (b)

Many responses did not make the reasoning sufficiently clear to gain this mark. Learners needed to state whether their comment referred to the actual motion with air resistance or to the motion predicted by the model in which air resistance was ignored.

Many responses stated that the acceleration in the model would be "different" from the acceleration in the actual motion. More complete responses explained that, with air resistance included, the actual acceleration, speed or velocity would be smaller.

Part (c)

Most learners gave a correct response to this part of the question. A significant number of responses, however, did not explain the effect that an extensible string would have "on the motion of the balls," as required by the question. Instead, some responses referred to the tension in the string and/or friction on the pulley.

Part (d)

Many responses gained the mark for finding the velocity of the balls at the instant when ball B hit the ground, but did not then go on to complete the method for the remaining marks. Of the responses that progressed further, most found the time taken for ball A to reach its maximum height. Fewer responses recognised that the same amount of time would be needed for the string to become taut again, so the time found needed to be doubled.

A significant minority of responses found the distance travelled in one second and then used this distance to find a time under gravity. This did not lead to a valid method for the question.